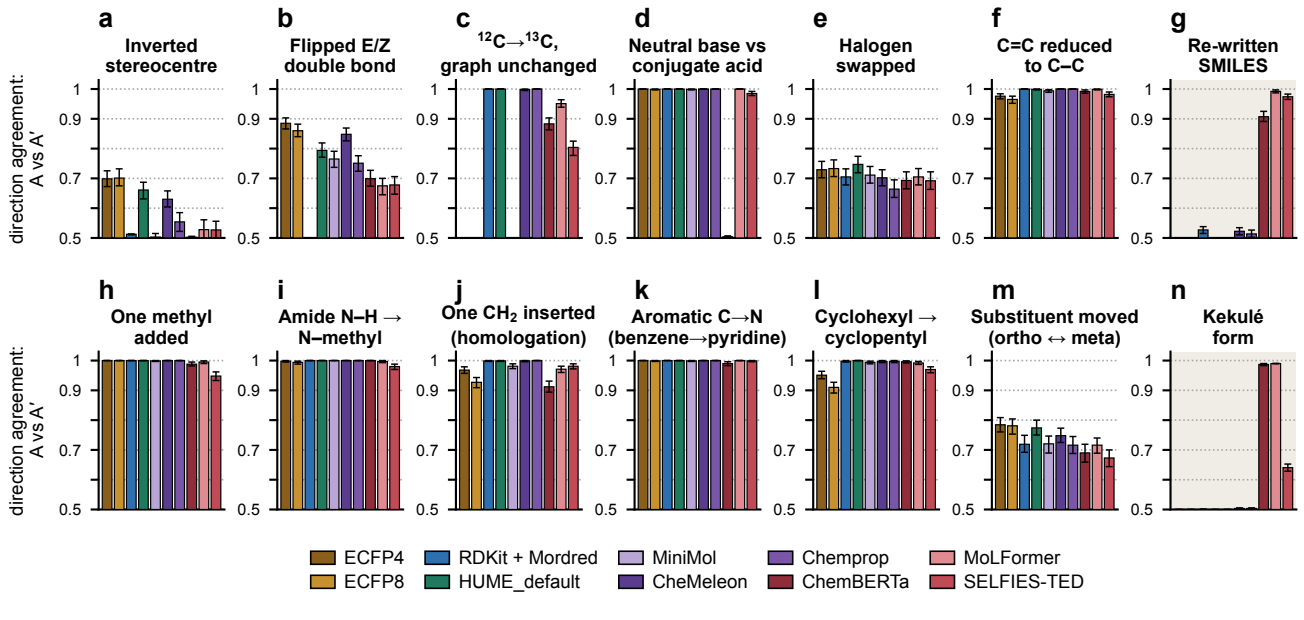

a
Inverted stereocentre
b
Flipped E/Z double bond
c
12C→13C, graph unchanged
d
Neutral base vs conjugate acid
e
Halogen swapped
f
C=C reduced to C–C
g
Re-written SMILES
h
One methyl added
i
Amide N–H → N–methyl
j
One CH2 inserted (homologation)
k
Aromatic C→N (benzene→pyridine)
l
Cyclohexyl → cyclopentyl
m
Substituent moved (ortho ↔ meta)
n
Kekulé form
direction agreement: A vs A'
1
0.9
0.7
0.5
ECFP4
ECFP8
RDKit + Mordred
HUME_default
MiniMol
CheMeleon
Chemprop
ChemBERTa
MoLFormer
SELFIES-TED